

# CMSC 320 Homework 3

October 20, 2024

## 1 Introduction

Throughout two semesters at the University of Maryland, College Park, students have participated in a survey that gathers some personal demographic information and their evaluations of human behavior in various real-life scenarios. This report employs statistical tests to establish correlations between students' survey responses regarding demographics, such as gender and political opinion, and their assessments of human behavior.

## 2 Background

The data includes 133 entries in the '2024' survey dataset, 144 in Dr. Alam's dataset labelled 'Fardina' from last semester, and 103 from Mr. Morawski's dataset labelled 'Max' from last semester, totaling 380 entries. These datasets contain some demographic questions, and questions that end with the phrase "Am I a jerk?", prompting the respondents to evaluate the actions of a human being in a set scenario, and picking one of three levels determining whether the person is a jerk or not. The '2024' dataset contains survey responses similar to the 'Fardina' and 'Max' datasets, but has the genders flipped in most of the questions regarding evaluation of human behavior. The 'Fardina' dataset also includes an additional question designed to perform priming on the survey participants. The following subsections detail the key issues presented by the datasets, and their solutions.

### 2.1 Data dropped at random

Some entries in the dataset had missing responses, either partially or completely. Given that these missing responses were random and represented only a small portion of the dataset, it was opted to drop these rows. Since the number of respondents was limited, a mode imputation was not appropriate.

### 2.2 Gender column data errors

The column that contained respondent's genders was titled "What bests represents your gender", which has been corrected to "What best represents your gender?". Additionally, a few entries in this column were labeled as 'Famale', which have been corrected to 'Female'.

### 2.3 College year data errors

All the Sophomore year student respondents were incorrectly labelled as 'Sophmore' in the data. It was corrected to reflect students' school year accurately.

### 2.4 Grammatical errors

Several columns contained grammatical errors in their titles. For instance, the phrase 'if this is how he treats herself' in the title of the 8th column of the 2024 dataset has been corrected to 'if this is how he treats himself,' as the person's gender can be inferred from the rest of the question. In the 'Fardina' dataset, one of the questions incorrectly ended with the phrase 'Am I jerk?' while the 'Max' dataset's

version of the question was grammatically correct. The phrase in the 'Fardina' dataset has been amended to ensure consistency across databases, especially while merging them.

## 2.5 Database Consistency

To facilitate merging databases, two columns have been adjusted. The age column in the '2024' dataset was initially populated with ages in decimal format, which has been converted to an integer format, aligning with the 'Max' and 'Fardina' databases. The 'Fardina' and 'Max' databases include a column titled 'How would you rate your religiousness / spirituality?' with three possible categories. The '2024' database features a similar column titled 'How would you rate your religiousness?' with the same categories/levels. Consequently, column title and values have been modified for the '2024' database to conform to the answer types of the former databases.

## 3 Analysis

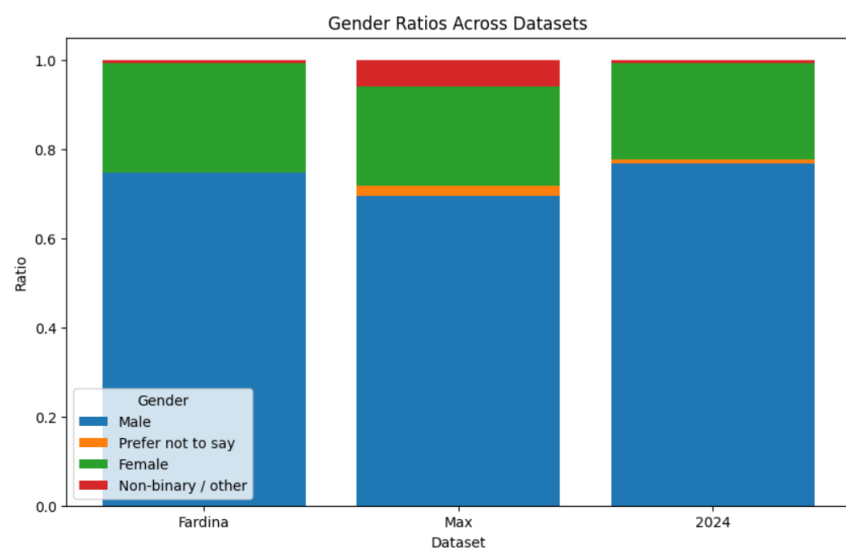
This dataset has been analyzed according to two main hypotheses-

- Does gender have a statistically significant effect on how questions pertaining to evaluation of human behavior are answered?
- Does political opinion have a statistically significant effect on how questions pertaining to evaluation of human behavior are answered?

The following subsections provide an in-depth study of these two hypotheses using the data available to us.

### 3.1 Does gender have a statistically significant effect on how questions pertaining to evaluation of human behavior are answered?

A few things to consider are the ratio of men to women in the responses, and which dataset(s) to use, as the '2024' dataset contains survey questions where the primary gender is male, while both other datasets contain questions where the primary gender is female. By graphing the gender of students who participated in the survey, we see that there is a high ratio of male to female respondents.

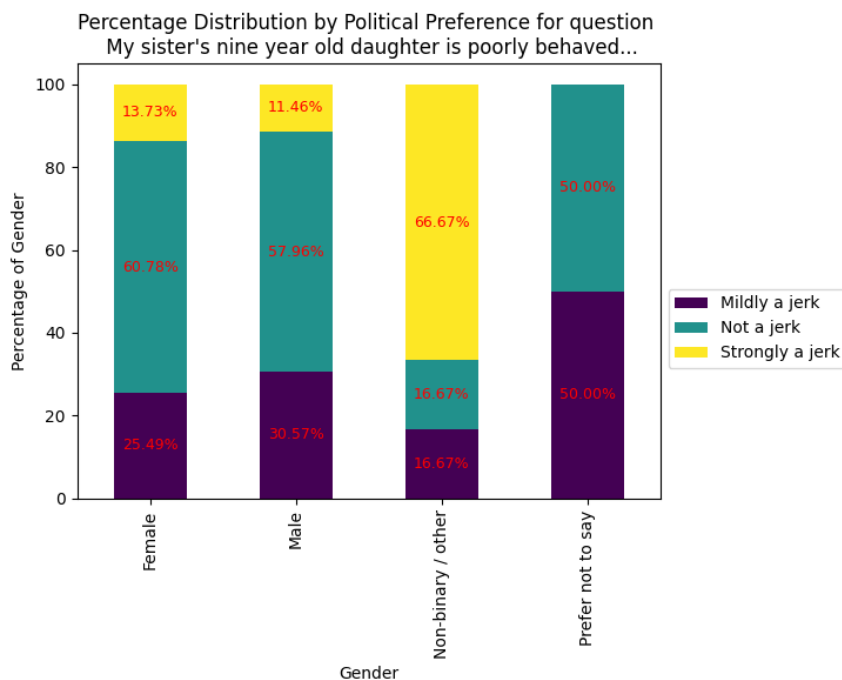


Based on this observation, it is appropriate to examine the correlation between gender and other survey questions as proportions of the total number of respondents for each gender. Given that all responses are categorical, chi-squared tests are employed to determine whether gender has any statistically significant impact on the survey responses using the null hypothesis that gender does have a statistically significant effect on responses to questions evaluating human behavior. The chi-squared test is performed separately on the combined ‘Max’ and ‘Fardina’ datasets, and the ‘2024’ dataset, to address the issue of flipped genders within the questions.

The chi-squared test reveals that there is a fair, statistically significant relationship between gender and one particular behavior evaluation question in the combined ‘Max’ and ‘Fardina’ datasets, with a p-value of 0.0126. The relevant question states-

*“My sister’s nine year old daughter is poorly behaved. One day, my sister dropped the daughter off on my doorstep without my consent and drove off, texting me that she needed me to watch her daughter for the day. The daughter complained of being hungry, so I gave her the leftovers I had. She said she didn't like them, but I told her that's all there was. She put them in the microwave until they burned, then told me she couldn't eat them and we would have to go out and get food. I told her that was what there was to eat, and if she ruined it she'd have to wait for her mom to get back. Later, everyone yelled at me for 'starving' a child. Am I a jerk?”*

To further understand this significance, a graph has been drawn to compare the responses of female and male participants to this answer using the combined dataset.



Looking at the data, one can infer that more women are either voting for the ‘Not a jerk’ / ‘Strongly a jerk’ categories, causing more polarization of opinions. Although very few in number, Non-Binary / other students voted more for the ‘Strongly a jerk’ category, and the 2 students who chose not to reveal their gender are divided between the ‘Not a jerk’ / ‘Mildly a jerk’ categories. However, the significance of this correlation is still relatively difficult to assume by a simple graph.

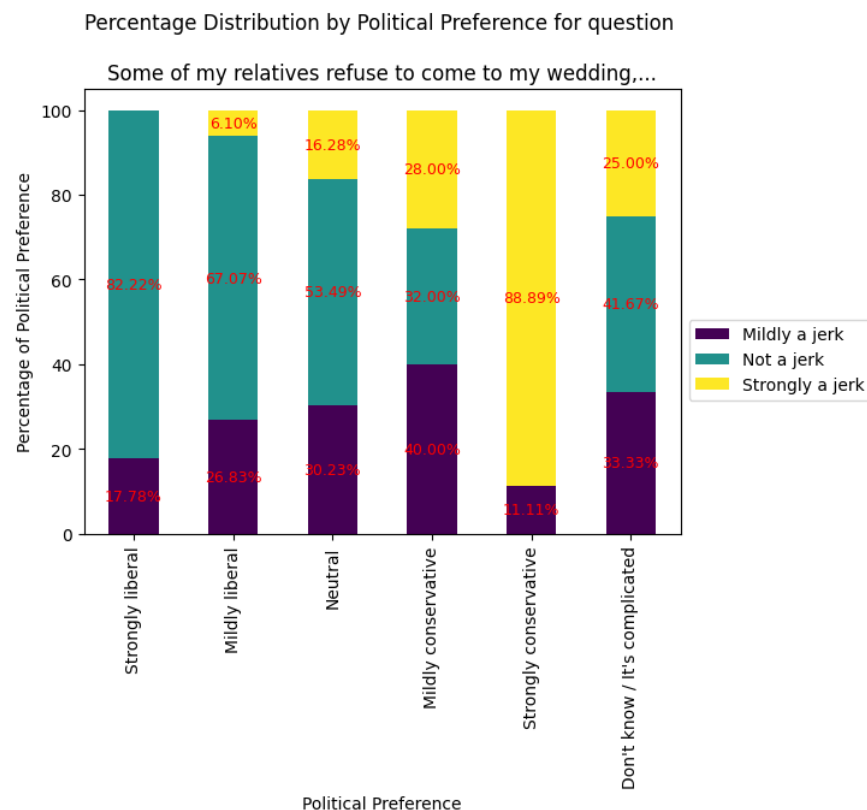
### 3.2 Does political opinion have a statistically significant effect on how questions pertaining to evaluation of human behavior are answered?

In line with the tests conducted to determine whether gender significantly impacts survey responses, a chi-squared statistical test and a stacked bar graph will be employed to assess the effect of political opinion on the same questions. The null hypothesis in this case will be whether political opinion has a statistically significant effect on how questions pertaining to the evaluation of human behavior are answered.

The chi-squared test reveals strong statistically significant relationships between political opinion and many questions in the dataset where the primary subject of the question is female. Conversely, the more recent '2024' dataset does not yield any significant results. Political opinion has an exceedingly strong statistical effect on one behavior evaluation question in the combined 'Max' and 'Fardina' datasets, with a p-value of  $5.5 \times 10^{-11}$ . The relevant question states-

*"Some of my relatives refuse to come to my wedding, since they don't approve of our 'lifestyle'. I would like to donate the money I will save to an LGBTQ organization in their name. I'm hoping that I or the organization will be able to send them a receipt/thank you for the donation. Would I be a jerk?"*

However, this observation is peculiar, as there is no specific gender mentioned in the question, meaning the question itself should be consistent across all three datasets. The '2024' dataset provides a p-value of 0.2 for this same question, indicating that the null hypothesis is not entirely correct. Nevertheless, a graph using only the 'Max' and 'Fardina' datasets can provide more insight into this strong p-value.



The graph vividly illustrates the trend from strongly liberal to strongly conservative respondents of ranking how much of a ‘jerk’ the person in the given scenario is. Most strongly liberal respondents believe the person is ‘not a jerk’, while most strongly conservative respondents believe the person is ‘strongly a jerk’. The categories ‘mildly liberal’, ‘neutral’ and ‘mildly conservative’ depict an increasing trend of people perceiving the person in question as ‘strongly a jerk’, and less likely ‘not a jerk’.

## 4 Conclusion

This study aimed to explore the impact of gender and political opinion on students' evaluations of human behavior through a survey conducted over two semesters at the University of Maryland, College Park. By employing chi-squared statistical tests and analyzing the data through various graphical representations, significant insights were uncovered.

The analysis revealed that gender has a statistically significant relationship with the evaluation of certain behaviors, particularly in scenarios where the primary subject is female. Similarly, political opinion demonstrated a strong influence on responses to specific questions, most notably in the combined ‘Max’ and ‘Fardina’ datasets. One behavior evaluation question showcased the polarization between strongly liberal and strongly conservative respondents, highlighting the influence of political leanings on moral judgments. The more recent '2024' dataset did not yield significant results when attempting to observe the significance of political opinions’ effect on the behavior evaluation questions, despite having one same question as the datasets from previous semesters. One can reasonably conclude that political opinion does not impact students’ perception of human behavior as significantly as it did from previous semesters. Further data collection could be resourceful to verifying this observation.